Contribution Journal: Lab 10 Unsupervised Learning

ITAI 1371 · Submitted November 7, 2026

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Group and members: JosephClay (individual: Joseph Clay)

1. Which tasks or cells did you personally write or debug?

- **Data setup (cell 3):** Ran the given `make_blobs` code and confirmed the scatter plot showed 300 points in four clear groups.
- **Task 1, Elbow Method (cell 5):** Wrote the loop for k = 1 to 10, fit KMeans at each k, stored `inertia_`, and plotted the elbow curve. Inertia dropped steeply up to k = 4 and flattened after.
- **Task 2, K-Means (cell 7):** Fit KMeans with k = 4, pulled `labels_` and `cluster_centers_`, and plotted the clusters by color with the centers marked as red X's.
- **Task 3, PCA (cell 10):** Reduced Iris from 4 features to 2 components, printed explained variance (PC1 92.46%, PC2 5.31%, total 97.77%), built the results DataFrame, and plotted it by species.
- **Debugging:** Task 3 failed with `NameError: name 'sns' is not defined` because the notebook never imports seaborn. Fixed it by adding `import seaborn as sns`.
- **Knowledge Check:** Wrote answers to all three questions.
- **Final run:** Removed a stray empty cell, re-ran the notebook, and confirmed every code cell has output.

2. What did each teammate contribute?

No other group members. I completed all tasks independently.

3. Commits or files pushed to GitHub

- Repository: https://github.com/ClayClimate-AI/itai-1371-lab10-unsupervised-learning
- Files: `L10_JosephClay_Notebook.ipynb`, `L10_JosephClay_Notebook.pdf`, `L10_JosephClay_Reflection.pdf`, `L10_JosephClay_Contribution.pdf`, `README.md`

4. Decisions you made or influenced

- **Working file:** The downloaded notebook raised a SyntaxError because its line breaks were missing, so I worked from a copy with only the line breaks restored.
- **Scaling:** Kept the blob data unscaled, following the notebook hints. In the cell 3 plot, both features cover similar ranges.

- **Choosing k:** Picked k = 4 from my own elbow plot. It matched the four groups visible in the raw data.
- **Checking the 95% claim:** Knowledge Check question 3 says two components explain "over 95%." I checked that against my own output: 97.77%.
- **Repository contents:** Pushed only the deliverables. Support files and the local Python environment are excluded.

5. Time spent and how the work was split

Completed in one working session on November 7, 2026. All work was done by me.

Tools: I ran the notebook locally in VS Code using a Python virtual environment. I used Claude Code (an AI assistant) to set up the environment, explain errors, check my outputs, format my written answers, and export the PDFs. I wrote the notebook code and the answers myself.